Background Summary: Text-Conditioned 3D Medical Localization

The Problem

This project sits at the intersection of medical image segmentation and vision language grounding.

In medical imaging, classification just answers the question of whether a tumor is present. Segmentation, or localization, answers a harder question: exactly which voxels belong to the tumor. A model can be very good at the first task while being quite poor at the second, and the two are often confused with each other in casual discussion of model performance.

Text conditioned grounding takes this a step further. Instead of predicting from a fixed set of output classes, the model is given a natural language description, something like "the enhancing tumor region," and has to find the matching region inside a 3D scan. This is the medical imaging version of vision language models like CLIP, which learn to match images and text by embedding both into a shared space so that matching pairs end up close together.

Small lesions turn out to be the hard case. Many efficient architectures compress an entire image or volume down into a single embedding vector. That compression tends to preserve just enough signal to detect that something is there, while losing the fine spatial detail needed to recover its precise boundary. This is a known failure mode in medical imaging, and the value of this project is that it quantifies the effect rigorously with statistics, rather than simply asserting that it happens.

The Methodology

The modeling approach borrows the contrastive alignment idea behind CLIP. An image and a piece of text are each turned into a vector, and the model is trained so that matching image text pairs end up with high cosine similarity while non matching pairs end up far apart. The foundational reference for this idea is Radford et al., "Learning Transferable Visual Models From Natural Language Supervision", commonly known as the CLIP paper.

The volume encoder is a 3D ResNet 10, adapted to work on volumetric patches instead of flat 2D images, built using MONAI, which is a widely used open source deep learning toolkit built specifically for medical imaging.

The text side uses PubMedBERT, a version of BERT that was pretrained on biomedical literature instead of general web text. It is used here in a frozen state, meaning its weights are not updated during training.

For localization, the usual tool people reach for is Grad-CAM, a technique for visualizing where inside an image a convolutional network is focusing its attention. It does not work here, because this architecture pools each patch down into a single vector, which destroys the spatial information Grad-CAM needs.

In place of Grad-CAM, the project slides a small window across the entire volume, scores each window position against the text embedding, and stitches those scores together into a heatmap.

Model quality at the localization step is measured with Dice and IoU, both standard overlap scores that compare a predicted tumor region against the region an expert actually annotated. A score of one means perfect overlap and a score of zero means no overlap at all.

Because the project runs many experiments and compares them against each other pairwise, it also leans on two statistical safeguards. The Wilcoxon signed-rank test is a non parametric test for paired differences that does not assume the data is normally distributed. Benjamini-Hochberg correction is used on top of that to control the false discovery rate, which matters once you are running many simultaneous statistical tests and want to avoid mistaking noise for a real effect.

The Dataset

The data comes from BraTS2020, a widely used and publicly available benchmark of brain MRI scans with expert annotated tumor regions, hosted by CBICA at the University of Pennsylvania. It is a standard dataset in medical image segmentation research. This project pulled its copy from the Kaggle mirror of BraTS2020.